

Exercise 1: Control Structures

Creating Table:

```
CREATE TABLE Customers (  
    CustomerID INT PRIMARY KEY,  
    CustomerName VARCHAR2(100),  
    Age INT,  
    Balance NUMBER(10, 2),  
    IsVIP VARCHAR2(10) DEFAULT 'FALSE'  
);
```

```
CREATE TABLE Loans (  
    LoanID INT PRIMARY KEY,  
    CustomerID INT,  
    InterestRate NUMBER(5, 2),  
    DueDate DATE,  
    CONSTRAINT fk_customer FOREIGN KEY (CustomerID) REFERENCES  
Customers(CustomerID)  
);
```

```
INSERT INTO Customers (CustomerID, CustomerName, Age, Balance) VALUES (1, 'Alice Smith',  
65, 5000);  
INSERT INTO Customers (CustomerID, CustomerName, Age, Balance) VALUES (2, 'Bob Jones',  
45, 15000);  
INSERT INTO Customers (CustomerID, CustomerName, Age, Balance) VALUES (3, 'Charlie  
Brown', 70, 12000);  
INSERT INTO Customers (CustomerID, CustomerName, Age, Balance) VALUES (4, 'Diana Prince',  
30, 2000);
```

```
INSERT INTO Loans (LoanID, CustomerID, InterestRate, DueDate) VALUES (101, 1, 5.5, SYSDATE  
+ 10);  
INSERT INTO Loans (LoanID, CustomerID, InterestRate, DueDate) VALUES (102, 2, 6.0, SYSDATE  
+ 45);  
INSERT INTO Loans (LoanID, CustomerID, InterestRate, DueDate) VALUES (103, 3, 4.5, SYSDATE  
+ 20);
```

```
COMMIT;
```

Scenario 1:

```
BEGIN
  FOR rec IN (SELECT L.LoanID, C.Age, L.InterestRate
               FROM Loans L
               JOIN Customers C ON L.CustomerID = C.CustomerID)
  LOOP
    IF rec.Age > 60 THEN
      UPDATE Loans
      SET InterestRate = InterestRate - 1
      WHERE LoanID = rec.LoanID;

      DBMS_OUTPUT.PUT_LINE('Discount applied to Loan ID: ' || rec.LoanID || '. New Rate: ' ||
                           (rec.InterestRate - 1));
    END IF;
  END LOOP;
  COMMIT;
END;
/
```

Scenario 2:

```
BEGIN
  FOR rec IN (SELECT CustomerID, Balance FROM Customers)
  LOOP
    IF rec.Balance > 10000 THEN
      UPDATE Customers
      SET IsVIP = 'TRUE'
      WHERE CustomerID = rec.CustomerID;

      DBMS_OUTPUT.PUT_LINE('Customer ' || rec.CustomerID || ' promoted to VIP.');
```

Scenario 3:

```
DECLARE
  CURSOR c_due_loans IS
    SELECT L.LoanID, C.CustomerName, L.DueDate
    FROM Loans L
    JOIN Customers C ON L.CustomerID = C.CustomerID
    WHERE L.DueDate BETWEEN SYSDATE AND (SYSDATE + 30);
BEGIN
  DBMS_OUTPUT.PUT_LINE('--- SENDING REMINDERS ---');
  FOR rec IN c_due_loans LOOP
    DBMS_OUTPUT.PUT_LINE('Reminder: ' || rec.CustomerName ||
      ', your loan ' || rec.LoanID ||
      ' is due on ' || TO_CHAR(rec.DueDate, 'DD-MON-YYYY'));
  END LOOP;
END;
/
```

Updated Table:

```
BEGIN
  DBMS_OUTPUT.PUT_LINE('-----');
  DBMS_OUTPUT.PUT_LINE('UPDATED CUSTOMERS TABLE');
  DBMS_OUTPUT.PUT_LINE('ID | Name      | Age | Balance | IsVIP');
  DBMS_OUTPUT.PUT_LINE('-----');

  FOR c IN (SELECT * FROM Customers ORDER BY CustomerID) LOOP
    DBMS_OUTPUT.PUT_LINE(
      RPAD(c.CustomerID, 3) || ' | ' ||
      RPAD(c.CustomerName, 14) || ' | ' ||
      RPAD(c.Age, 3) || ' | ' ||
      RPAD(c.Balance, 8) || ' | ' ||
      c.IsVIP
    );
  END LOOP;

  DBMS_OUTPUT.PUT_LINE(CHR(10));

  DBMS_OUTPUT.PUT_LINE('-----');
  DBMS_OUTPUT.PUT_LINE('UPDATED LOANS TABLE');
  DBMS_OUTPUT.PUT_LINE('LoanID | CustID | Interest Rate | Due Date');
  DBMS_OUTPUT.PUT_LINE('-----');

  FOR l IN (SELECT * FROM Loans ORDER BY LoanID) LOOP
    DBMS_OUTPUT.PUT_LINE(
      RPAD(l.LoanID, 6) || ' | ' ||
      RPAD(l.CustomerID, 6) || ' | ' ||
      RPAD(l.InterestRate, 13) || ' | ' ||
      TO_CHAR(l.DueDate, 'DD-MON-YYYY')
    );
  END LOOP;
```

```

);
END LOOP;
DBMS_OUTPUT.PUT_LINE('-----');
END;
/

```

Output:

Output:

Discount applied to Loan ID: 101. New Rate: 4.5

Discount applied to Loan ID: 103. New Rate: 3.5

Customer 2 promoted to VIP.

Customer 3 promoted to VIP.

--- SENDING REMINDERS ---

Reminder: Alice Smith, your loan 101 is due on 03-JUL-2026

Reminder: Charlie Brown, your loan 103 is due on 13-JUL-2026

UPDATED CUSTOMERS TABLE

ID	Name	Age	Balance	IsVIP
----	------	-----	---------	-------

1	Alice Smith	65	5000	FALSE
2	Bob Jones	45	15000	TRUE
3	Charlie Brown	70	12000	TRUE
4	Diana Prince	30	2000	FALSE

UPDATED LOANS TABLE

LoanID	CustID	Interest Rate	Due Date
--------	--------	---------------	----------

101	1	4.5	03-JUL-2026
102	2	6	07-AUG-2026
103	3	3.5	13-JUL-2026
